

ECO387C - Fall 2024

Problem Set 6

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You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.¹

The representative households solve

$$\max_{c_t} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} \quad (1)$$

subject to

$$\begin{aligned} & c_t + b_t + q_t s_t + p_t^c o_t^c + p_t^p o_t^p \\ = & b_{t-1} (1 + r_{t-1}) + [q_t + d_t] s_{t-1} \\ & + (q_t + d_t - \bar{p}^c) I \{q_t + d_t - \bar{p}^c \geq 0\} o_{t-1}^c \\ & + (\bar{p}^p - q_t - d_t) I \{\bar{p}^p - q_t - d_t \geq 0\} o_{t-1}^p \end{aligned}$$

and the starting conditions $b_{-1} = o_{-1}^c = o_{-1}^p = 0$ and $s_{-1} = 1$. $I(\cdot)$ denotes the indicator function. The notation is as follows.

1. b_t denotes the regular bond holdings. If the agent buys a bond in period $t-1$, she is entitled to principal plus interest $(1 + r_{t-1})$ in period t .
2. s_t are the stock holdings, d_t the associated dividend and q_t the price.
3. o_t^c denotes a one-period call option which gives the holder the right to purchase one share at price $\bar{p}^c$ (the strike or exercise price). We will assume that the option has to be exercised *before* dividends are paid.
4. o_t^p denotes a one-period put option which gives the holder the right to sell one share at price $\bar{p}^p$. Again, we will assume that the option has to be exercised *before* dividends are paid.

Questions

¹Please note that this problem set is due on 11:59PM, October 30th 2024.

1. Set up the Lagrangian and derive the asset pricing relationships

$$0 = \mathbb{E}_t [m_{t,t+1}(1+r_t) - 1] \quad (2)$$

$$0 = \mathbb{E}_t \left[m_{t,t+1} \frac{q_{t+1} + d_{t+1}}{q_t} - 1 \right] \quad (3)$$

$$0 = \mathbb{E}_t \left[m_{t,t+1} \frac{q_{t+1} + d_{t+1} - \bar{p}^c}{p_t^c} I \{q_{t+1} + d_{t+1} - \bar{p}^c \geq 0\} - 1 \right] \quad (4)$$

$$0 = \mathbb{E}_t \left[m_{t,t+1} \frac{\bar{p}^p - q_{t+1} - d_{t+1}}{p_t^p} I \{\bar{p}^p - q_{t+1} - d_{t+1} \geq 0\} - 1 \right] \quad (5)$$

where

$$m_{t,t+1} = \beta \left(\frac{c_t}{c_{t+1}} \right)^{\frac{1}{\sigma}}$$

is the stochastic discount factor.

2. Download the dataset `GMMdata.mat` from Canvas, copy it into the MATLAB folder you are working in and load the dataset using the command `load GMMdata`. This dataset contains time series on d_t , c_t , r_t , q_t , p_t^c , and p_t^p as shown in Table 1.

t	d_t	c_t	r_t	q_t	p_t^c	p_t^p
1	2.62	2.62	-0.26	161.1	42.5	61.0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1000	2.37	2.37	0.11	106.8	28.2	10.7

Table 1: Structure of the dataset

3. **OLS ESTIMATION:** Run Hall's regression to estimate the intertemporal elasticity of substitution.

$$\ln(c_{t+1}) - \ln(c_t) = \alpha + \sigma r_t + \varepsilon_t$$

where ε_t is a rational expectations error.

4. **GMM ESTIMATION:** The parameters $\bar{p}^c, \bar{p}^p$ are given by 100 and 150, respectively, but β, σ are unknown. The goal is to estimate these by GMM. For simplicity, we will not use the optimal weighting matrix, but just the identity matrix. To set up the objective function, replace the expectations in (2) to (5) with their sample analogue and write

$$G(\sigma, \beta) = \begin{pmatrix} G_1 \\ G_2 \\ G_3 \\ G_4 \end{pmatrix} = \begin{pmatrix} \frac{1}{T-1} \sum_{t=1}^{T-1} [m_{t,t+1}(1+r_t) - 1] \\ \frac{1}{T-1} \sum_{t=1}^{T-1} \left[m_{t,t+1} \frac{q_{t+1} + d_{t+1}}{q_t} - 1 \right] \\ \frac{1}{T-1} \sum_{t=1}^{T-1} \left[m_{t,t+1} \frac{q_{t+1} + d_{t+1} - \bar{p}^c}{p_t^c} I \{q_{t+1} + d_{t+1} - \bar{p}^c \geq 0\} - 1 \right] \\ \frac{1}{T-1} \sum_{t=1}^{T-1} \left[m_{t,t+1} \frac{\bar{p}^p - q_{t+1} - d_{t+1}}{p_t^p} I \{\bar{p}^p - q_{t+1} - d_{t+1} \geq 0\} - 1 \right] \end{pmatrix}. \quad (6)$$

Now the GMM estimator is defined as

$$\begin{pmatrix} \hat{\sigma} \\ \hat{\beta} \end{pmatrix} = \arg \min_{\sigma, \beta} G'G.$$

- (a) Estimate σ and β by GMM using the full vector $G = (G_1, G_2, G_3, G_4)'$. (Hint: Set up a separate function in MATLAB which computes $G'G$. Then use `fminsearch` to minimize this objective.)
- (b) Now drop G_4 from (6) and redo the estimation. Do the estimates change? Next drop G_3 and G_4 and redo the estimation. Finally, estimate σ and β from $G = G_1$ only. What happens to the estimates? Explain.